Human reproduction II: Birth

An embryo obtains food and oxygen from the mother's blood via the *placenta*, which also removes waste. The placenta forms on the wall of the uterus as the embryo develops. The placenta and embryo are joined by the *umbilical cord*. The blood of the mother and baby do not meet or mix. The two blood supplies are separate.

1 Will a baby necessarily have the same blood group as the mother? Explain your answer.



The baby floats in a bag of liquid called the *amniotic fluid*. The fluid acts as a shock absorber and protects the baby. It cushions the baby from minor physical knocks the mother may receive. Samples of the amniotic fluid containing cells of the unborn baby can be used to check the sex of the child or for early detection of certain disorders.

After about nine months the baby is usually fully developed and in the correct position to be born. Hormones within the mother's body signal the uterus to begin contractions, or *labour*. These contractions start to push the baby out.

The baby passes head-first, usually face-down, through the cervix and down the vaginal canal. After the baby is born, the umbilical cord is tied and cut. The navel, or belly button, is where the cord was attached. There are no nerves in the cord, so it does not hurt the baby when the cord is severed. The placenta (or afterbirth) comes away from the mother soon after the baby is born. It is checked for abnormalities and weighed (it should weigh about the same as the baby, otherwise it may not have been fully expelled). It is then discarded, unless permission has been given to use it for medical purposes, such as the extraction of stem cells.

Some babies are not in the upside-down position when they are ready to be born. These babies are either born bottom first in a breech birth, which can cause injury to both mother and baby, or by caesarean section. A caesarean section involves surgically removing the baby by cutting through the mother's abdominal wall, while the mother is under a general anaesthetic.

2 Describe the function of an anaesthetic.

3 Give another name for the navel.

4 What causes contractions in the uterus to start?

Identify the terms described by the following definitions:

5 *Cord coming from the navel connecting the foetus with the placenta.*

6 *Organ that gives food and oxygen to foetus and removes carbon dioxide and waste.*

7 *Delivery of a baby by cutting mother's abdomen and uterus.*

8 *Narrow, outer end of uterus.*

9 *The fluid in which the embryo is suspended within the amnion.*

10 *Organ of the female mammal that holds developing embryo.*

11 *Process of giving birth.*

Research

12 Outline precautions that pregnant women should take to avoid harm to their unborn child.

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13 Use words from box to fill in the spaces.

AMNIOTIC	AFTERBIRTH	ANIMAL	BABY	CELLS	CERVIX	CYTOPLASM
DIVIDE	EMBRYO	FERTILISATION	GROWS	IMPLANTATION	LIFE	MICROSCOPE
NAVEL	NUCLEUS	ORGANISM	OVARIES	OVIDUCT	OVUM	OXYGEN
PENIS	PLACENTA	SPERM	TESTES	UMBILICAL	UTERUS	ZYGOTE

All living things are made up of _____. Cells are the building blocks of _____ . The simplest form of life is a unicellular animal, or _____, with one cell. You need a _____ to see cells properly. Plant cells and _____ cells are different. The _____ controls all the activities of the cell. The cell membrane holds the contents of the cell and the _____ is the material in the cell apart from the nucleus.

Male sex cells are called _____ and female sex cells are called _____ (plural ova). When a sperm joins with an ovum, _____ takes place forming a new, single cell called a _____. Ova (eggs) are produced by the _____ of a female and pass through the _____ . Sperm cells are produced in the _____ (testicles) and are passed out of the _____ during sexual intercourse when the penis is placed inside the vagina. Sperm move through the _____ (or womb) and, if an ovum is present, fertilisation can take place.

The zygote cells then _____ repeatedly and it moves towards the uterus. It becomes embedded on the wall of the uterus – this is called _____. The fertilised cells develop further to become an _____, then a foetus and eventually to be born as a baby. As the embryo _____ in the womb it is nourished and protected. It is surrounded by a sac filled with _____ fluid. During pregnancy an organ called the _____ develops. This allows food and _____ to go to the developing foetus and takes away waste products. The placenta extends into the _____ cord. The umbilical cord is a cord of blood vessels joining the _____ of the foetus to the placenta.

Approximately nine months after fertilisation, the baby is born. The uterus wall contracts, the amniotic fluid, in which the baby has been floating, escapes and the _____ is pushed through the _____ and vagina. The umbilical cord is cut, the placenta is expelled as the _____ and the baby starts to use its own lungs.

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